ERGA Assembly Report

v24.10.15

Tags: ERGA-Community

TxID	3733116
ToLID	**ddWikPhil2**
Species	Wikstroemia phillyreifolia
Class	Magnoliopsida
Order	Malvales

Genome Traits	Expected	Observed
Haploid size (bp)	909,417,095	990,784,725
Haploid Number	9 (source: direct)	9
Ploidy	2 (source: ancestor)	2
Sample Sex	XX	XX

EBP metrics summary and curation notes

Obtained EBP quality metric for hap1: 8.8.Q69
Obtained EBP quality metric for hap2: 7.8.Q73

The following metrics were automatically flagged as below EBP recommended standards or different from expected:

. Kmer completeness value is less than 90 for hap1
. Kmer completeness value is less than 90 for hap2

Curator notes

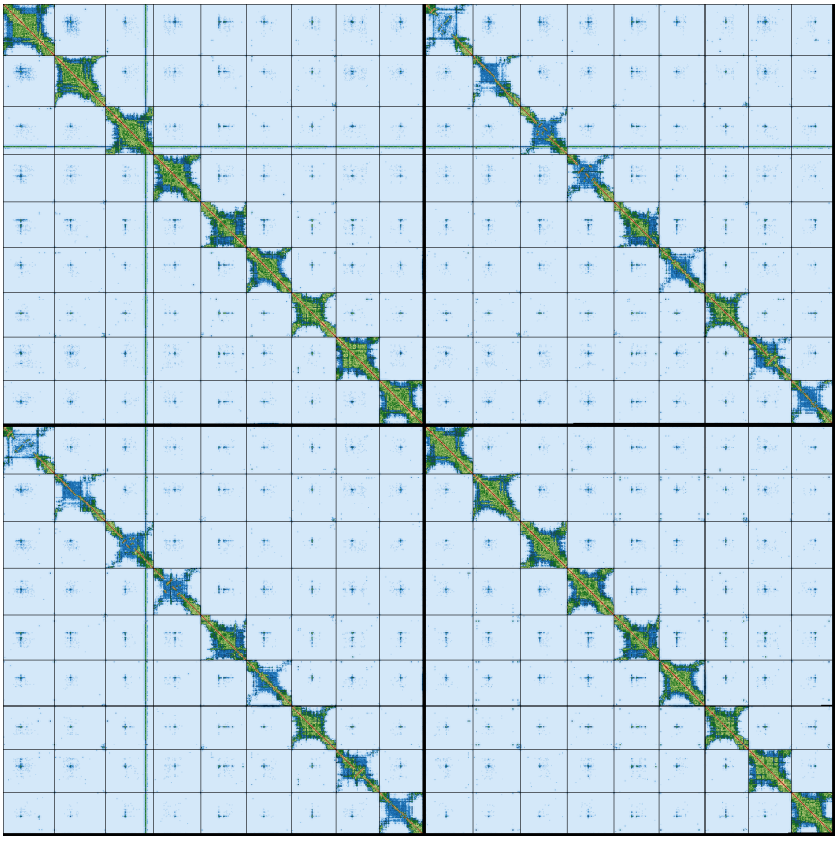
. Interventions/Gb: NA
. Contamination notes: "It is likely that plastid and mitochondrial scaffolds are present."
. Other observations: "The contact maps contains both haplotypes. Hap2 has been named and oriented to match hap1. Sex chromosomes are yet to be determined."

Quality metrics table

Metrics	Pre-curation hap1	Pre-curation hap2	Curated hap1	Curated hap2
Total bp	977,194,348	973,973,832	990,784,725	960,328,559
GC %	41.17	41.17	41.2	41.14
Gaps/Gbp	10.23	10.27	7.07	13.54
Total gap bp	1,000	1,000	700	1,300
Scaffolds	172	65	161	74
Scaffold N50	106,850,007	106,666,668	106,850,007	106,414,849
Scaffold L50	5	5	5	5
Scaffold L90	9	9	9	9
Contigs	182	75	168	87
Contig N50	101,553,079	99,901,047	106,850,007	83,790,665
Contig L50	5	5	5	5
Contig L90	12	11	10	13
QV	69.1	73.7	69.1	73.7
Kmer compl.	79.27	79.58	79.83	79.01
BUSCO sing.	95.5%	95.7%	95.7%	95.5%
BUSCO dupl.	1.9%	1.8%	1.9%	1.8%
BUSCO frag.	0.7%	0.6%	0.6%	0.7%
BUSCO miss.	1.9%	1.8%	1.8%	1.9%

BUSCO: 6.1.0 (euk_genome_min, miniprot) / Lineage: eudicotyledons_odb12 (genomes:76, BUSCOs:2805)

HiC contact map of curated assembly

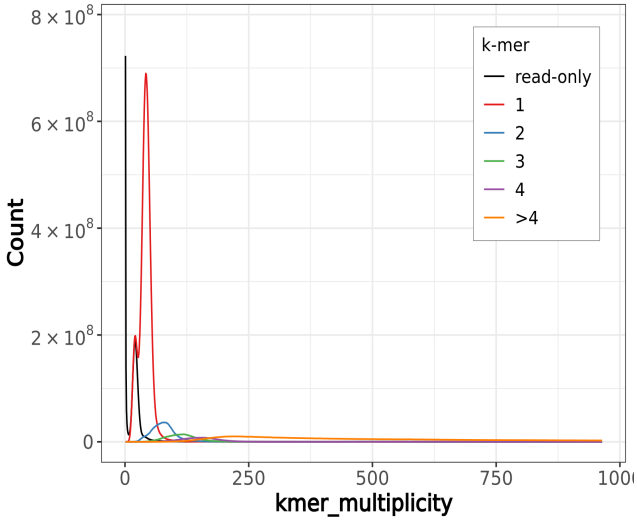


hap1 [LINK]

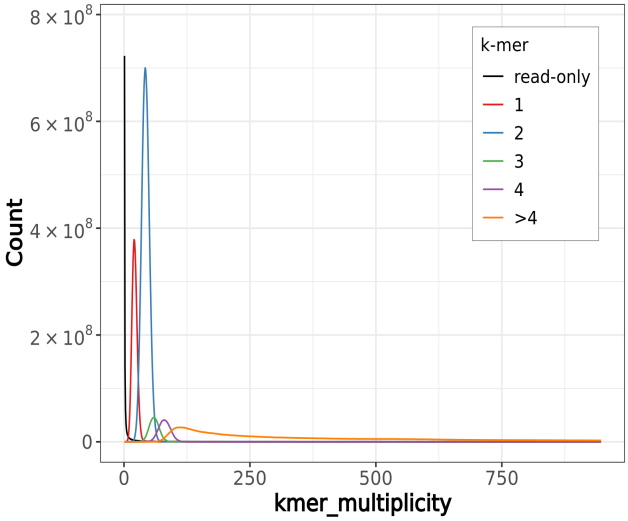
hap2 HiC PNG is missing!

hap2 File link is missing!

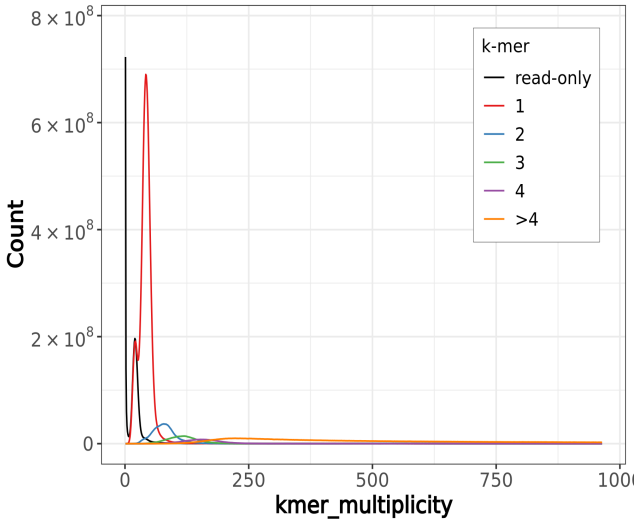
K-mer spectra of curated assembly



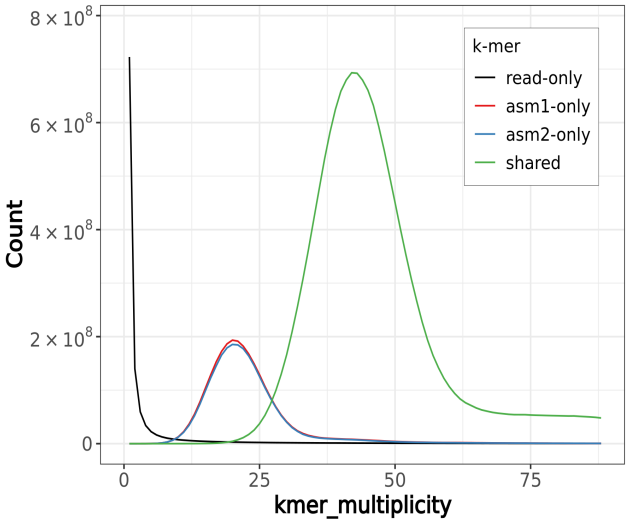
Distribution of k-mer counts per copy numbers found in **asm1** (hapl.)



Distribution of k-mer counts per copy numbers found in asm (dipl.)

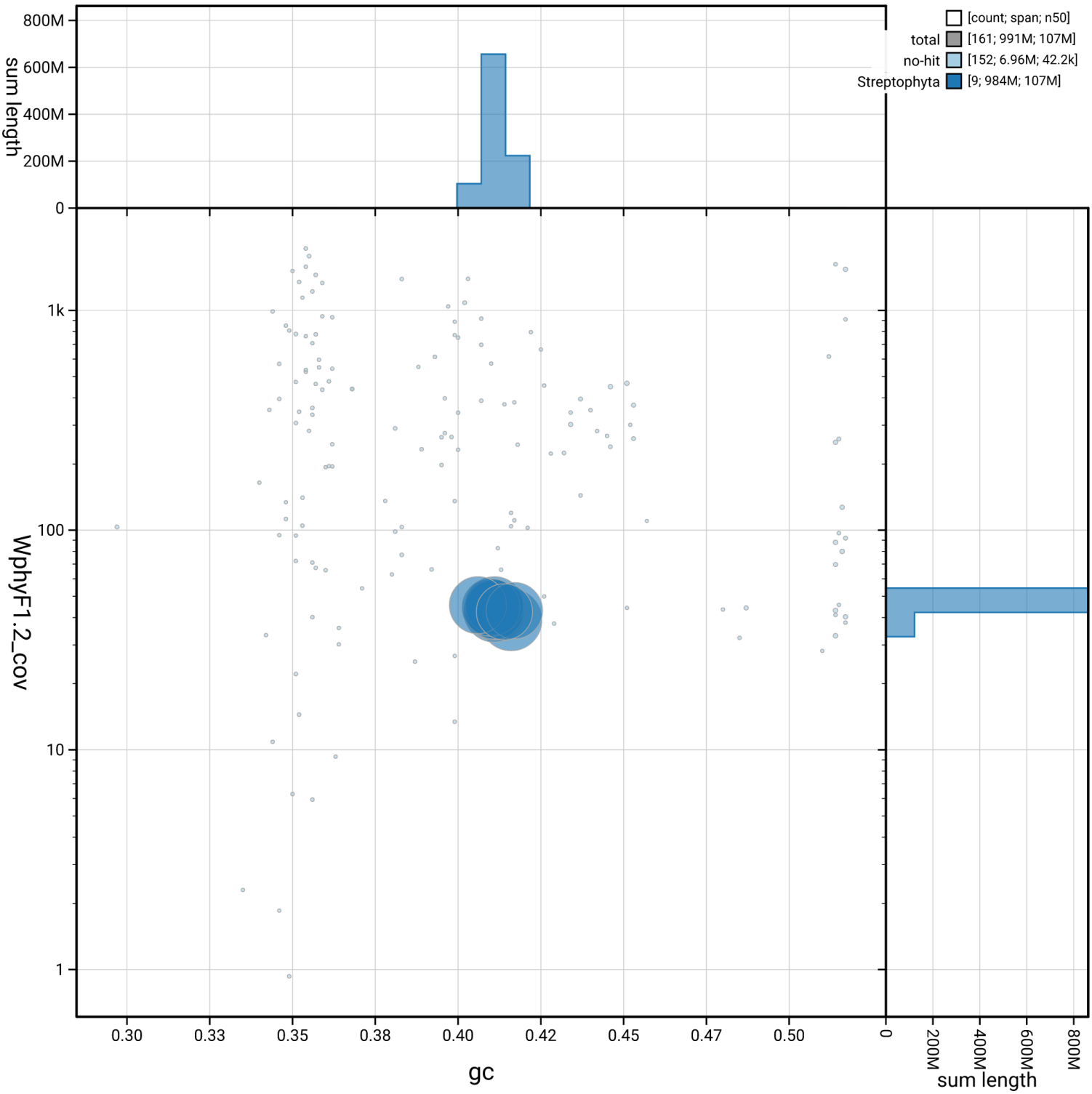


Distribution of k-mer counts per copy numbers found in **asm2** (hapl.)

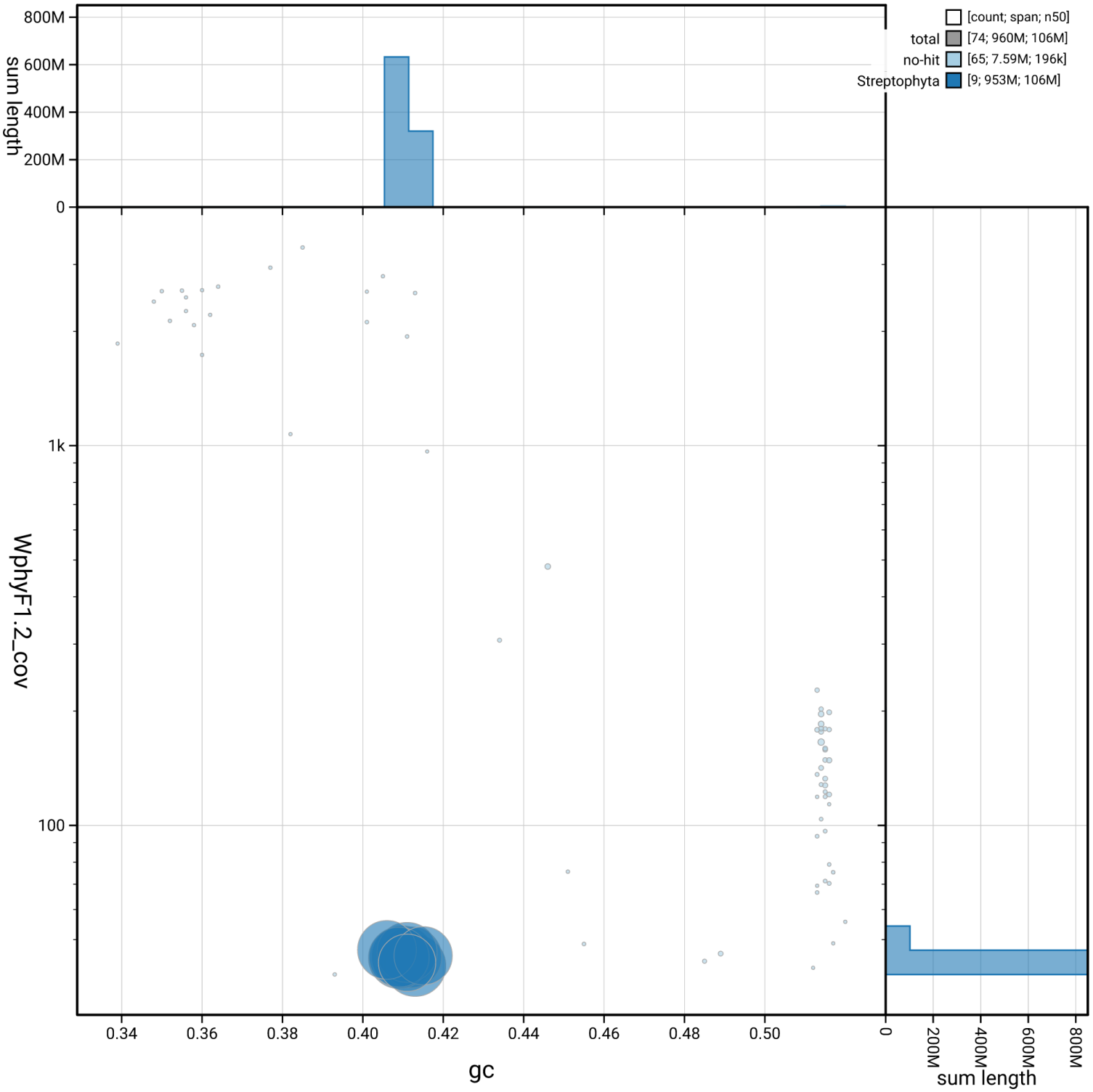


Distribution of k-mer counts coloured by their presence in reads/assemblies

Post-curation contamination screening



hap1. Bubble plot circles are scaled by sequence length, positioned by coverage and GC proportion, and coloured by taxonomy. Histograms show total assembly length distribution on each axis.



hap2. Bubble plot circles are scaled by sequence length, positioned by coverage and GC proportion, and coloured by taxonomy. Histograms show total assembly length distribution on each axis.

Data profile

Data	HiFi	Hi-C
Coverage	21	42

Assembly pipeline

- **sanger-tol/genomeassembly**
  - |_ *ver:* b8153da
  - |_ *key param:* assembler: hifiasm
  - |_ *key param:* long_read_platform: pacbio_hifi
  - |_ *key param:* purge: false
  - |_ *key param:* phased_assembly: true
  - |_ *key param:* busco_lineage: eudicotyledons_odb12
  - |_ *key param:* find_mito: false

Curation pipeline

- **PretextViewAI**
  - |_ *ver:* 1.0.7
  - |_ *key param:* NA
- **apg-tpf-utils**
  - |_ *ver:* ebb67fe
  - |_ *key param:* NA
- **sanger-tol/treeval**
  - |_ *ver:* 1.4.7
  - |_ *key param:* mode: FULL_COMBINED
  - |_ *key param:* run_ultra: true
  - |_ *key param:* assembly: assem_level:scaffold defined_class: plant
  - |_ *key param:* assem_reads: read_type: hifi
  - |_ *key param:* hic_data: hic_aligner: bwamem2
  - |_ *key param:* kmer_profile: kmer_length: 31
  - |_ *key param:* telomere: teloseq: TTAGGG
  - |_ *key param:* busco: lineage: eudicotyledons_odb12
  - |_ *key param:* map_order: length
- **FastGA**
  - |_ *ver:* 1.5
  - |_ *key param:* NA
- **sanger-tol/curation-resources**
  - |_ *ver:* b4da326
  - |_ *key param:* NA
- **GEP2**
  - |_ *ver:* 0.7
  - |_ *key param:* NA
- **nikostr/blobtoolkit**
  - |_ *ver:* aa1d476
  - |_ *key param:* align: true

Submitter: Nikos Tsardakas Renhuldt
Affiliation: Lund University

Date and time: 2026-10-12 14:18:39 CEST